LESSON PLAN

COMPOSITE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	11	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Surface roughness, meaning of real surface, geometrical surface, effective surface, and various irregularities.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Surface roughness, meaning of real surface, geometrical surface, effective surface, and various irregularities.
5. Specific Objectives	19.1.1.1 Trainee will be able to describe details about Engineering Metrology.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of surface finish types using magnified samples; Safety: Handle samples carefully; Supervision: Instructor monitors handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of metrology in engineering?
(b) Can you list three types of surface irregularities?
(c) Why is surface finish critical for aircraft components?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of geometrical surface • Concept of real surface • Meaning of effective surface • Distinction between ideal and actual surfaces	L & Ds	PPT, IB	15 min
(b)	• Classification of surface irregularities • Long wavelength irregularities • Short wavelength irregularities • Impact of roughness on performance	L & Ds	PPT, PDF	20 min
(c)	• Measurement of surface roughness • Standard roughness parameters • Practical implications in maintenance • Comparison of finish types	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between geometrical and real surfaces.
(b) How do long wavelength irregularities affect component fit?
(c) Why is effective surface important in metrology?

12. Home Assignment and Follow Up	Prepare a brief note on how surface roughness affects the sealing efficiency of aircraft joints, referencing LP 02 maintenance procedures.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Standard room maintenance procedure, inspection, standard maintenance and safety precaution	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Wing Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min